

SOCIAL MEDIA AND WEB ANALYTICS

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INTRODUCTION TO THE ASSIGNMENT

GOAL

- The main goal of this assignment is to **predict a performance indicator** from a certain sector by using social media and web data, however the assignment itself is split off into a **descriptive** and **predictive** part
- You are free to choose your own sector:
 - Car sales
 - Cosmetic brand sales
 - Company sales fixtures
 - Box office sales
 - (Crypto currency) stock prices
 - ...
- You should **scrape** the social media and web **data** and the dependent variable **yourself**
- Remember that there are **limits** to the number of posts you can scrape. So, you might want to scrape regularly to have all the data points you need.

PART 1: DESCRIPTIVE

- Gather the posts **regularly** using your hashtags/mentions/subreddits of choice
- You should not limit yourself to platforms discussed in class, there are tons of APIs available that can provide additional data sources
- Describe with appropriate **visuals** what you notice over the time
- You can use several techniques to summarize the posts:
 - Wordclouds
 - Topic modelling
 - Word embeddings
 - Word networks
 - Sentiment analysis
 - ...
- **The goal is to get insight, so there is no right or wrong!**

PART 1: DESCRIPTIVE

- Try to give insight into the **network** and identify **network effects** on Bsky or Mastodon
- For example, identification of **influencers** in the network
 - Valuable to know which users are at the center of the network (e.g., high degree/betweenness) and how they are situated in the network
 - Targeting these influencers can be valuable since they control the flow of information and spread the message easily
 - See *Lecture1 Homework.ipynb*
 - Second degree follower network build for 3 users in the example
 - Shows you the limits in extraction possibilities from social media, so resort to assumed relationships
- Note for Mastodon
 - You can only scrape the full followers list of an account if your account is on the same server (e.g., mastodon.social).
 - If you are on different servers, you can only see the followers of the profile that are on your server.

PART 2: PREDICTIVE

- Decide upon the proper **experimental design**
 - Time-based split to create the dependent variable
 - Train/test split to assess performance evaluation
- Perform **feature engineering**: text and sentiment analysis on the social media messages
- Incorporate **additional** web data **sources** besides social media data (optional)
- **Build** and **interpret** the predictive models: discuss model performance, the most important variables, and their relationship with the dependent

PART 2: PREDICTIVE

- Focus on **social media analytics**, not only prediction
 - The goal is **not** to build the most accurate predictive model but to apply the right **social media analytical techniques** (e.g., text representation).
- Follow the correct **workflow**
 - The emphasis is on **performing each step correctly and in the right order**, rather than purely optimizing model performance.
- There is no one-size-fits-all modeling approach
 - You are free to use **any predictive model** covered in this course or from your data mining/machine learning courses.

2. PERFORMANCE

- Some **examples** of projects (feel free to be creative and think of your own idea)
 - Predict box office sales of upcoming movies
(<https://www.movieinsider.com/movies/march/2024>)
 - Scrape box offices sales daily (<https://www.the-numbers.com/daily-box-office-chart>)
 - Scrape interesting fora about movies or specific producers like marvel studios
 - Scrape critic and audience movie scores and reviews (e.g., imdb, rottentomatoes)
 - Predict (crypto) stock prices
 - Use a stock market API to scrape the daily/hourly prices of the currencies/companies
 - Scrape interesting fora about bitcoin or specific companies (e.g., samsung)
 - Scrape other relevant economic data
 - Belgian government: <https://statbel.fgov.be/>
 - US Government: <https://www.census.gov/data/developers/guidance/api-user-guide.html>
 - Company annual accounts: <https://www.nbb.be/nl/balanscentrale>

DELIVERABLES

- **Presentation**
 - 20 minutes: 12 minutes for the two projects, 8 minutes Q&A
 - Present your business rationale, methodology, and findings
 - Be creative !
- **Technical report**
 - Max 5 pages, appendices not included and 12pt
 - A concise, but in-depth explanation of your methodology and references to academic literature to underpin your chosen methods
- **Code:** a link to your UGent-Github included in the report
- Both the presentation and technical report account for **55% of your grades (with peer evaluation)**
- **Deadline:** Presentation and code should be submitted on via Ufora (naming: Presentation_GroupX.ppt and Report_GroupX.pdf)
 - Deadline DS4B: **Thursday 2 April**
 - Deadline HIR & double degree: **Tuesday 21 April**
- **Defense on MS Teams:**
 - Defense DS4B: **Friday 3 April**
 - Defense HIR & double degree: **Week of 20 April**
 - Timeslots will be made available on Ufora